causalmetrics lecture note, Section 4

Doubly Robust Estimation and Double Machine Learning

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Contents

1 The problem

1.1 Same identification, a harder estimation problem

Section 03 ended with five estimators for one estimand. All five assumed the same thing: conditional ignorability, $(Y(1), Y(0)) \perp\!\!\!\perp D \mid X$, plus overlap. They differed only in *how* they used two objects. This lecture calls them the **nuisance functions**,

$$m(X) = \mathbb{E}[D \mid X], \qquad g(d, X) = \mathbb{E}[Y \mid D = d, X],$$

together with their residual-based cousin $\ell(X) = \mathbb{E}[Y \mid X]$. Regression adjustment imputes with g. Weighting rebalances with m. AIPW uses both. In Section 03 these functions were quadratic polynomials in one covariate. That was enough to teach the geometry. It is not enough for the data this course is about.

Two features of modern applied work break the Section 03 toolkit at the estimation stage, not at the identification stage.

- **Many covariates.** A customer record carries purchase histories at many horizons, engagement counts, geography, device, and tenure. A firm-level panel has dozens of balance-sheet ratios. Text and image embeddings add hundreds of coordinates. With p comparable to n, least squares on everything is noisy or impossible. Choosing controls by hand brings back the discretion that a pre-specified adjustment set was meant to remove.
- **Unknown functional form.** Even with few covariates, nothing says the outcome or the assignment rule is linear in them. Targeting systems, eligibility rules, and habit formation produce thresholds, curvature, and interactions. A misspecified linear adjustment is biased. No standard error reports that bias.

The instinct of anyone who has fitted a random forest is obvious: *let the machine learn g and m.* Prediction is what these methods are good at. Both nuisance functions are conditional expectations, so they are prediction targets. This lecture follows that instinct. It shows why the obvious way of doing it fails, and what the repair looks like. The repair is called **double/debiased machine learning**, DML for short.

One sentence to hold on to: **prediction is the means, not the goal.** The target is a single number θ, a causal effect, together with a confidence interval that has its nominal coverage. Every choice below is judged by that standard, not by out-of-sample fit.

1.1.1 Notation

We observe $W = (Y, D, X)$. Two working models carry the lecture.

- The **partially linear model** (PLM):

$$Y = \theta D + g(X) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D, X] = 0; \qquad D = m(X) + v, \quad \mathbb{E}[v \mid X] = 0.$$

  The effect is a constant θ; D may be binary or continuous; g and m are unrestricted.
- The **interactive regression model** (IRM): $Y = g(D, X) + \varepsilon$ with binary D and no restriction on how D and X interact. The target is the ATE, $\theta = \mathbb{E}[g(1, X) - g(0, X)]$, or the ATT.

In both models θ is a **low-dimensional target** and the functions (g, m) or (ℓ, m) are **infinite-dimensional nuisances**. Under conditional ignorability the PLM coefficient and the IRM contrast are causal effects; without it they are still well-defined predictive quantities, and DML estimates them either way. The identification argument is the one from Section 03 and is not repeated here.

1.2 The natural approach: plug a learner into last lecture's formula

Take the imputation view of Section 03 and replace the linear model with a forest. Fit one random forest of Y on (D, X), predict every unit twice, once with D set to 1 and once with D set to 0, and average the difference:

$$\hat{\theta}_{\text{plug-in}} = \frac{1}{n} \sum_{i=1}^{n} \{\hat{g}(1, X_i) - \hat{g}(0, X_i)\}.$$

This is regression adjustment with a flexible learner (an "S-learner" in the heterogeneous-effects literature). It is tempting for three reasons.

- Forests are consistent for g under mild conditions.
- No functional form is assumed.
- No propensity model is needed.

A naive analyst reports the standard deviation of the imputed differences divided by $\sqrt{n}$, as for any sample mean.

```
dat <- simulate_dml_dgp(n = 500, p = 10, theta = 1, seed = 1)
x_names <- paste0("x", 1:10)

set.seed(1)
naive_plugin_forest(dat, x_names, num.trees = 100)
#>  estimate std.error
#> 0.6739098 0.0226867
```

The truth is 1. The estimate is far below it, and the reported standard error would not have warned you. Whether this is bad luck or a systematic failure is a question for a simulation.

1.3 One experiment, six estimators

1.3.1 The design

Ten standard normal covariates; three of them matter. The propensity score and the outcome surface are nonlinear in the *same* covariates. So the confounding is real, and a linear adjustment is misspecified:

$$m_0(X) = \Lambda(0.8X_1 - 0.4(X_2^2 - 1)), \qquad g_0(X) = 2\,\Lambda(2X_1) + 0.5X_2^2 + 0.25X_3,$$

$$D \sim \text{Bernoulli}(m_0(X)), \qquad Y = \theta D + g_0(X) + \varepsilon, \qquad \varepsilon \sim N(0,1), \qquad \theta = 1,$$

where Λ is the logistic function. The effect is constant, so the ATE, the ATT, and the partially linear coefficient coincide and every estimator below targets the same number. Each replication draws $n = 500$ units; there are 100 replications.

1.3.2 The estimators

Estimator	Nuisances	Fitted on	What it stands for
OLS with linear controls	none	full sample	Section 03's regression adjustment
Naive plug-in forest	$\hat{g}(D, X)$	full sample	"replace the linear model by a forest"
Naive plug-in forest, cross-fitted	$\hat{g}(D, X)$	held-out folds	the wrong score with the right sample splitting
Orthogonal score, full-sample forests	$\hat{\ell}(X), \hat{m}(X)$	full sample	the right score with the wrong sample use
Orthogonal score, cross-fitted	$\hat{\ell}(X), \hat{m}(X)$	held-out folds	DML with the partially linear score
AIPW, cross-fitted	$\hat{m}(X), \hat{g}(0, X), \hat{g}(1, X)$	held-out folds	Section 03's AIPW with forests: DML in the interactive model

The last three are `est_dml()` calls. The cross-fitted plug-in is there to separate two ideas that are easy to conflate: sample splitting and the choice of score. "Orthogonal score" means the residual-on-residual regression of Section 03's Series R: predict Y and D from X, then regress $Y - \hat{\ell}(X)$ on $D - \hat{m}(X)$. Why that combination deserves the name is the subject of Section 2. "Cross-fitted" means each unit's nuisance predictions come from forests that never saw that unit. That is the recipe of Section 03's Appendix C and the subject of Section 4. All forests use 100 trees and default tuning. The point is not to tune well. It is to see what tuning cannot fix.

```
results <- run_dml_experiment(
  n_sims = n_sims, n = 500, p = 10, theta = 1,
  num.trees = 100, folds = 5, seed = 1
)
tab <- summarise_dml_experiment(results)
pick <- function(estimator, column) tab[[column]][tab$estimator == estimator]
```

Table 1: Monte Carlo comparison, true effect 1, n = 500. Coverage is the share of replications whose nominal 95 percent interval contains the truth.

Estimator	Bias	SD	RMSE	Mean reported SE	95% coverage
OLS with linear controls	-0.320	0.124	0.343	0.119	0.23
Naive plug-in forest	-0.333	0.103	0.348	0.022	0.00
Naive plug-in forest, cross-fitted	-0.360	0.098	0.373	0.013	0.00
Orthogonal score, full-sample forests	-0.287	0.073	0.297	0.083	0.04
Orthogonal score, cross-fitted	-0.095	0.096	0.134	0.104	0.86
AIPW, cross-fitted	-0.087	0.106	0.136	0.116	0.90

```
plot_dml_experiment(results)
```

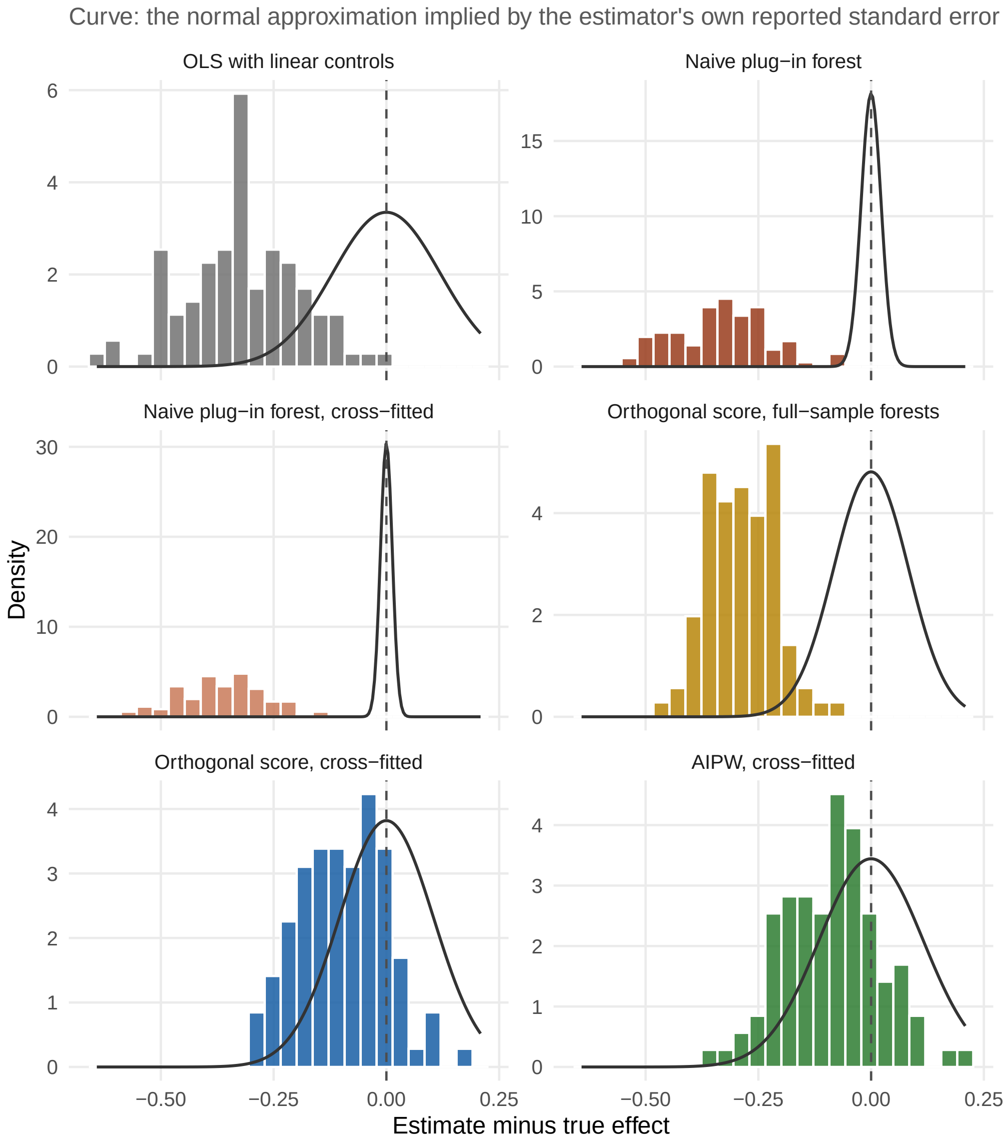

Figure 1: Sampling distributions of the estimators, centred at the truth. The curve in each panel is the normal approximation implied by the estimator's own average reported standard error: a histogram that is shifted relative to its curve is biased, one that is wider than its curve under-reports uncertainty.

1.3.3 Reading the results

- **Linear adjustment fails as expected.** OLS with all ten covariates has a bias of -0.32 and coverage of 0.23. The functional form is wrong, and more covariates would not help. This is the problem the forest was supposed to solve.
- **The plug-in forest is worse, not better.** Its bias, -0.33, is of the same size as OLS's, and its reported standard error, 0.022, is a fraction of the actual dispersion, 0.103, so the interval almost never covers the truth. The forest shrinks the treatment contrast. At every split D competes against ten covariates. Splits on X_1 and X_2 predict Y almost as well, so they absorb most of the effect of D. A learner tuned for prediction has no reason to protect the coefficient we care about.
- **Sample splitting alone does not help.** The cross-fitted plug-in keeps a bias of -0.36, and its reported standard error is just as misleading. The moment it solves is the wrong one. Cross-fitting cures a different disease.

- **The right score with the wrong sample splitting is still biased.** Fitting the orthogonal score's forests on the full sample gives a bias of -0.29 and coverage of 0.04. The residuals $Y - \hat{\ell}$ and $D - \hat{m}$ are too small and are correlated with the noise they were fitted to.
- **Cross-fitted DML recovers the target.** Both cross-fitted estimators are centred close to the truth (bias -0.10 and -0.09), their reported standard errors match the actual dispersion, and their coverage is close to nominal. The AIPW row is the estimator of Section 03 with forest nuisances: cross-fitted AIPW *is* DML for the ATE.
- **A small gap remains.** The cross-fitted rows are not exactly centred. That residual is nuisance estimation error leaking into $\hat{\theta}$ at second order. Section 2.4 shows why it shrinks with the accuracy of the learners, not with the size of the sample alone.

1.4 What went wrong: two biases

The plug-in estimator fails for two separate reasons, and it is worth keeping them apart because they have separate fixes.

1.4.1 Regularization bias

Machine learning estimators of a conditional expectation are biased by construction. Shrinkage, pruning, minimum leaf sizes, early stopping, and penalties all trade bias for variance. The resulting error $\hat{g} - g_0$ vanishes more slowly than $n^{-1/2}$. That is harmless for prediction. It is fatal for the plug-in, because the plug-in moment condition,

$$M(\theta, g) = \mathbb{E}[g(1, X) - g(0, X)] - \theta,$$

moves one-for-one with the nuisance. Perturb g_0 in a direction Δ. The moment moves by $\mathbb{E}[\Delta(1, X) - \Delta(0, X)]$, which is not zero in general. The naive partially linear moment $\mathbb{E}[(Y - \theta D - g(X))\, D] = 0$ has the same flaw; its derivative in the direction Δ is $-\mathbb{E}[\Delta(X)D]$. So the learner's bias in $\hat{g}$ passes undiminished to $\hat{\theta}$. That bias shrinks slower than the standard error, so the confidence interval never catches up. The repair is a moment whose derivative with respect to every nuisance is zero at the truth. That property is **Neyman orthogonality**, the subject of Section 2. The residual-on-residual score and the AIPW score both have it.

1.4.2 Overfitting bias

Orthogonality removes the *first-order* effect of nuisance error. It does not help when the nuisance estimate is correlated with the observations on which the score is evaluated. A forest fitted on the full sample reproduces part of each unit's noise ε_i in its own prediction $\hat{g}(X_i)$. The residual $Y_i - \hat{g}(X_i)$ then contains too little of ε_i. The residual $D_i - \hat{m}(X_i)$ contains too little of v_i. The score averages their product, which is biased. The more flexible the learner, the worse this "own-observation" bias. **Cross-fitting** breaks the dependence: the nuisances are fitted on folds that exclude the unit being scored. Section 4 shows why this is enough for the estimator to behave as if the true nuisances had been plugged in.

1.4.3 The remaining gap

With an orthogonal score and cross-fitting, nuisance error enters $\hat{\theta}$ only at second order. The remainders are products such as $\|\hat{\ell} - \ell_0\| \cdot \|\hat{m} - m_0\|$ and, for the partialling-out score, $\|\hat{m} - m_0\|^2$. They vanish at the $\sqrt{n}$ rate only if each nuisance converges faster than $n^{-1/4}$. In the experiment the forest's propensity error is close to that threshold at $n = 500$. That is why the cross-fitted rows still show a small negative bias. Better learners, more data, or a score that is robust to the harder nuisance all shrink it. Section 2.4 quantifies this, and Section 5 turns it into a workflow for choosing learners.

1.5 When this is the right tool

DML is an estimation layer. It sits on top of an identification strategy and changes nothing about it. Reach for it when all four conditions hold.

- **The design is credible without it.** Conditional ignorability with a pre-treatment adjustment set (this lecture). Or another design whose estimator contains nuisance functions: instrumental variables (Section 05), difference-in-differences (Section 07), policy learning (Section 08).
- **The nuisance functions are hard.** Many covariates, unknown functional forms, interactions, or engineered features such as embeddings. With three covariates and a plausible linear model, Section 03's tools are simpler and equally valid.
- **The target is low-dimensional.** An ATE, an ATT, a partially linear coefficient, a group-average effect, or a small vector of such numbers. DML delivers $\sqrt{n}$ inference for these. It does not deliver a nonparametric function with the same guarantees.
- **There is enough data to learn the nuisances.** The rate condition is a real constraint. A few hundred observations and a forest on fifty covariates will not satisfy it. The residual bias above will then be large.

Do not expect it to do four other things. It does not choose the adjustment set; a bad control stays bad. It does not repair a lack of overlap; extreme propensities remain extreme, and trimming still changes the estimand. It does not fix unobserved confounding. It does not estimate heterogeneous effects for individual units. The typical applications in this course fit the four conditions: targeted promotions with rich customer histories (Ellickson, Kar, and Reeder 2023), retirement-plan eligibility with income and demographics (Chernozhukov et al. 2018), and price elasticities with high-dimensional product features.

1.6 Doubly robust, double machine learning

The two names in this lecture's title are related but not synonyms.

- **Doubly robust** is a property of a *score*. The AIPW score of Section 03 is consistent for the ATE if either the outcome model or the propensity model is correct. Its bias is the product of the two errors.
- **Double machine learning** is a *recipe* with three ingredients: a Neyman-orthogonal score, cross-fitted nuisance estimation, and learners that converge fast enough. The AIPW score is one orthogonal score. Cross-fitted AIPW with machine-learned nuisances is DML for the ATE. The residual-on-residual score of the partially linear model is another orthogonal score. It is *not* doubly robust: its bias is second order in the propensity error alone.

The rest of the lecture develops the recipe.

- Section 2: orthogonality, the proof skeleton, and the rate condition.
- Section 3: three views of the score. The partially linear model as a generalized Frisch-Waugh-Lovell theorem, the interactive model with the doubly robust score, and double lasso, where the story started.
- Section 4: cross-fitting.
- Section 5: the recipe as a workflow. Folds, pooled versus fold-averaged solutions, repeated splits, learner selection.
- Section 6: diagnostics and failure modes.
- Section 7: the replications and a checklist.
- Section 8: where the recipe goes next.

2 Just enough framework

2.1 Targets, nuisances, moments

Everything in this lecture has the same shape.

- A **target** θ_0, a number or a short vector.
- A **nuisance** η_0, a function or a tuple of functions.
- A **score** $\psi(W; \theta, \eta)$ with moment

$$M(\theta, \eta) := \mathbb{E}[\psi(W; \theta, \eta)], \qquad M(\theta_0, \eta_0) = 0,$$

  and θ_0 is the unique root at the true nuisance.

The estimator replaces η_0 by a learned $\hat\eta$ and $\mathbb{E}$ by the sample mean $\mathbb{E}_n$, then solves $\mathbb{E}_n[\psi(W;\hat\theta,\hat\eta)] = 0$. The whole question is how the error $\hat\eta - \eta_0$ enters $\hat\theta$.

Three scores from Sections 1 and 03, with $H(D,X) = D/m(X) - (1-D)/(1-m(X))$:

Score	Nuisance η	$\psi(W;\theta,\eta)$
Plug-in (regression imputation)	g	$g(1,X) - g(0,X) - \theta$
Partialling out (PLM)	(ℓ, m)	$(Y - \ell(X) - \theta\,(D - m(X)))\,(D - m(X))$
AIPW (IRM)	(g, m)	$g(1,X) - g(0,X) + H(D,X)\,(Y - g(D,X)) - \theta$

2.2 Neyman orthogonality

Fix a direction $\Delta = \eta - \eta_0$. The **directional (Gateaux) derivative** of the moment at the truth is

$$\partial_\eta M(\theta_0,\eta_0)[\Delta] \;:=\; \frac{d}{dt}\, M(\theta_0, \eta_0 + t\Delta)\Big|_{t=0}.$$

> **Definition (Neyman orthogonality).** The score is Neyman orthogonal if $\partial_\eta M(\theta_0,\eta_0)[\Delta] = 0$ for every admissible direction Δ.

In words: a small error in the nuisance has no first-order effect on the moment. Only second-order terms survive. Check the three scores; one line each, full derivations in Appendix 9.1.

- **Plug-in.** $\partial_\eta M[\Delta] = \mathbb{E}[\Delta(1,X) - \Delta(0,X)]$. Not zero. Not orthogonal.
- **Partialling out.** Direction Δ_ℓ in ℓ: $-\mathbb{E}[\Delta_\ell(X)\,(D - m_0(X))] = -\mathbb{E}[\Delta_\ell(X)\,\mathbb{E}[D - m_0(X) \mid X]] = 0$. Direction Δ_m in m: the derivative is $-\mathbb{E}[\Delta_m(X)\,U] + \theta_0\,\mathbb{E}[\Delta_m(X)(D - m_0(X))]$ with $U = Y - \ell_0(X) - \theta_0(D - m_0(X))$; both terms vanish because $\mathbb{E}[U \mid D, X] = 0$ and $\mathbb{E}[D - m_0 \mid X] = 0$. Orthogonal.
- **AIPW.** Direction Δ_g in g: $\mathbb{E}[\Delta_g(1,X) - \Delta_g(0,X)] - \mathbb{E}[H_0(D,X)\,\Delta_g(D,X)] = 0$, because $\mathbb{E}[H_0\,\Delta_g(D,X)] = \mathbb{E}[\Delta_g(1,X)] - \mathbb{E}[\Delta_g(0,X)]$ (the inverse-propensity identity). Direction Δ_m in m: the derivative multiplies $Y - g_0(D,X)$, whose conditional mean is zero. Orthogonal.

The first bullet is Section 1's regularization bias in one line. The other two are the reason the orthogonal estimators recovered the target.

2.3 Why orthogonality is enough: skeleton of the proof

Take a score that is linear in θ, $\psi = \psi^b(W;\eta) - \psi^a(W;\eta)\,\theta$. All three scores above have this form. Then

$$\hat\theta = \frac{\mathbb{E}_n[\psi^b(W;\hat\eta)]}{\mathbb{E}_n[\psi^a(W;\hat\eta)]}, \qquad \sqrt{n}\,(\hat\theta - \theta_0) = \frac{\sqrt{n}\;\mathbb{E}_n[\psi(W;\theta_0,\hat\eta)]}{\mathbb{E}_n[\psi^a(W;\hat\eta)]}.$$

The denominator converges to $J_0 = \mathbb{E}[\psi^a(W;\eta_0)]$ once $\hat\eta$ is consistent. Everything happens in the numerator. Add and subtract the oracle score $\psi(W;\theta_0,\eta_0)$:

$$\sqrt{n}\,\mathbb{E}_n[\psi(\theta_0,\hat\eta)] = \underbrace{\sqrt{n}\,\mathbb{E}_n[\psi(\theta_0,\eta_0)]}_{(a)\ \text{oracle}} + \underbrace{\sqrt{n}\,(\mathbb{E}_n - \mathbb{E})[\psi(\theta_0,\hat\eta) - \psi(\theta_0,\eta_0)]}_{(b)\ \text{empirical process}} + \underbrace{\sqrt{n}\,\mathbb{E}[\psi(\theta_0,\hat\eta) - \psi(\theta_0,\eta_0)]}_{(c)\ \text{bias}},$$

where $\mathbb{E}$ in (b) and (c) treats $\hat\eta$ as fixed.

Step 1: the oracle term (a)**.** An average of i.i.d. mean-zero terms. By the central limit theorem, $(a) \to N(0, \mathbb{E}[\psi(W;\theta_0,\eta_0)^2])$. This is what an estimator with known nuisances would deliver.

Step 2: the bias term (c)**.** Expand $M(\theta_0,\cdot)$ around η_0:

$$(c) = \sqrt{n}\Big(\underbrace{\partial_\eta M(\theta_0,\eta_0)[\hat\eta - \eta_0]}_{=0\ \text{by orthogonality}} + \tfrac{1}{2}\,\partial^2_\eta M(\theta_0,\bar\eta)[\hat\eta - \eta_0, \hat\eta - \eta_0]\Big).$$

Orthogonality removes the first-order term. The second-order term is bounded by $C\,\|\hat\eta - \eta_0\|^2$, or by a product of the component errors. So $(c) = o_p(1)$ if $\|\hat\eta - \eta_0\| = o_p(n^{-1/4})$. That is the **rate condition**.

Step 3: the empirical process term (b). This is a centred average of a function that depends on $\hat\eta$, which was fitted on the same data. Without sample splitting it vanishes only if the learner lives in a "not too complex" (Donsker) class. Adaptive learners such as forests and boosting do not. **Cross-fitting** makes $\hat\eta$ independent of the fold on which it is evaluated. Conditional on the training folds, (b) has mean zero and variance $\mathbb{E}[(\psi(\hat\eta) - \psi(\eta_0))^2]$, which vanishes as soon as $\hat\eta$ is consistent. Section 4 gives the argument.

Conclusion.

$$\sqrt{n}\,(\hat\theta - \theta_0) \;\to\; N(0, V), \qquad V = \frac{\mathbb{E}[\psi(W;\theta_0,\eta_0)^2]}{J_0^2},$$

and V is estimated by plugging $\hat\eta$ and $\hat\theta$ into the same formula. Appendix 9.2 states the conditions and fills in the steps.

> **The three ingredients, mapped to the proof.** An orthogonal score kills the first-order part of (c). The rate condition kills the second-order part of (c). Cross-fitting kills (b). Drop any one and the corresponding term reappears.

Two consequences matter in practice.

- **Oracle equivalence.** $\hat\theta$ behaves as if the true nuisances had been plugged in. Standard errors are computed from the score with $\hat\eta$ inside, and no correction for the first stage is needed.
- **Learner agnosticism.** Nothing in the argument names the learner. Any method with the rate works, and different nuisances may use different methods.

2.4 The rate condition and double robustness

The second-order remainder of Step 2 has a concrete form for each score (Appendix 9.1).

- **Partialling out.**
$$\mathbb{E}[\psi(\theta_0,\hat\eta)] = \mathbb{E}[(\hat\ell - \ell_0)(\hat m - m_0)] - \theta_0\,\mathbb{E}[(\hat m - m_0)^2].$$
The bias of $\hat\theta$ is this quantity divided by $J_0 = \mathbb{E}[(D - m_0)^2]$. The rate condition is $\|\hat\ell - \ell_0\|\,\|\hat m - m_0\| + \|\hat m - m_0\|^2 = o(n^{-1/2})$. The propensity error enters *squared* and on its own. Nothing protects against a poor $\hat m$.
- **AIPW.**
$$\mathbb{E}[\psi(\theta_0,\hat\eta)] = \mathbb{E}\Big[\tfrac{\hat m - m_0}{\hat m}\,(\hat g(1,X) - g_0(1,X))\Big] + \mathbb{E}\Big[\tfrac{\hat m - m_0}{1-\hat m}\,(\hat g(0,X) - g_0(0,X))\Big].$$
Every term is a *product* of a propensity error and an outcome error. The rate condition is $\|\hat m - m_0\|\,\|\hat g - g_0\| = o(n^{-1/2})$. If either factor is zero the remainder is zero: that is **double robustness**, now seen as a special structure of the remainder rather than a separate property.

Section 1's residual bias is this remainder at work. Two cross-fitted fits on the design of Section 1, at $n = 500$ and $n = 2000$, show the propensity error against the $n^{-1/4}$ threshold and the bias it implies through the partialling-out formula.

```r
rate_check <- function(n, seed) {
  dat <- simulate_dml_dgp(n = n, p = 10, theta = 1, seed = seed)
  set.seed(seed)
  fit <- est_dml(
    dat, y = "y", d = "d", x = x_names,
    learner_l = lrn("regr.ranger", num.trees = 100),
    learner_m = lrn("classif.ranger", predict_type = "prob", num.trees = 100),
    folds = 5, seed = seed
  )
  rmse_m <- sqrt(mean((fit$nuisance$m_hat - dat$m_true)^2))
  j0 <- fit$diagnostics$identification$mean_sq_treatment_residual
  c(n = n, rmse_m = rmse_m, threshold = n^(-1/4),
    implied_bias = -1 * rmse_m^2 / j0, estimate = fit$estimate)
```

```
}
round(rbind(rate_check(500, 1), rate_check(2000, 1)), 3)
#>         n rmse_m threshold implied_bias estimate
#> [1,]  500  0.146     0.211       -0.088    0.943
#> [2,] 2000  0.119     0.150       -0.063    0.914
```

At $n = 500$ the forest's propensity error is of the same order as $n^{-1/4}$, and the implied bias matches the Monte Carlo bias of -0.10 in Table 1 in sign and size. At $n = 2000$ the error is smaller, but it shrinks slowly, because a forest's rate is well below $n^{-1/2}$. Two lessons follow. The learner for m deserves as much care as the learner for ℓ. And when one nuisance is much harder than the other, a doubly robust score, which only needs the product to be small, is the safer choice.

3 Three views of the score

The orthogonal score appears in three settings. Each gives a different picture of the same idea. Each subsection follows the same order: *model, theorem or score, estimator, R, when it breaks.*

3.1 Partially linear model: generalized Frisch-Waugh-Lovell

Model.

$$Y = \theta D + g(X) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D, X] = 0; \qquad D = m(X) + v, \quad \mathbb{E}[v \mid X] = 0.$$

D may be binary or continuous. Only θ is restricted to be a constant. Define $\ell(X) = \mathbb{E}[Y \mid X] = \theta\, m(X) + g(X)$ and the residuals $\tilde{Y} = Y - \ell(X)$, $\tilde{D} = D - m(X)$. Subtracting $\ell(X)$ from both sides of the model gives

$$\tilde{Y} = \theta\, \tilde{D} + \varepsilon.$$

This is Section 03's Series R, now as a statement about the population.

Theorem (partialling out in the PLM). If $\mathbb{E}[\tilde{D}^2] > 0$,

$$\theta = \frac{\mathbb{E}[\tilde{D}\tilde{Y}]}{\mathbb{E}[\tilde{D}^2]}.$$

Proof. Multiply $\tilde{Y} = \theta\tilde{D} + \varepsilon$ by $\tilde{D}$ and take expectations. The error term drops out because $\mathbb{E}[\varepsilon\tilde{D}] = \mathbb{E}\big[\mathbb{E}[\varepsilon \mid D, X]\,(D - m(X))\big] = 0$. □

With linear ℓ and m this is the Frisch-Waugh-Lovell theorem of Section 02. The nuisances are now arbitrary functions, and the residuals must be learned.

Score and estimator. The score $\psi = (\tilde{Y} - \theta\tilde{D})\tilde{D}$ is orthogonal (Section 2.2). With cross-fitted residuals $\check{Y}_i = Y_i - \hat{\ell}_{[k(i)]}(X_i)$ and $\check{D}_i = D_i - \hat{m}_{[k(i)]}(X_i)$,

$$\hat{\theta} = \frac{\sum_i \check{D}_i \check{Y}_i}{\sum_i \check{D}_i^2}, \qquad \widehat{\mathrm{SE}}(\hat{\theta}) = \frac{\sqrt{\sum_i \check{D}_i^2\, \hat{e}_i^2/(n-1)}}{\sum_i \check{D}_i^2/\sqrt{n}}, \qquad \hat{e}_i = \check{Y}_i - \hat{\theta}\check{D}_i.$$

The standard error is the heteroskedasticity-robust sandwich of the residual-on-residual regression. Nothing is added for the first stage; that is oracle equivalence.

Algorithm (DML for the partially linear model). 1. Split the indices into K folds $I_1, \dots, I_K$. 2. For each k, fit $\hat{\ell}_{[k]}$ and $\hat{m}_{[k]}$ on the data outside I_k; form $\check{Y}_i$ and $\check{D}_i$ for $i \in I_k$. 3. Regress $\check{Y}$ on $\check{D}$ by least squares; report the robust standard error.

In R. `est_dml(model = "plr")` runs the three steps. Two checks tie it to what you already know. With linear-regression nuisances fitted on the full sample, it *is* least squares with the controls included (classical FWL). The learners must be requested, because for a binary treatment the default learner for m is a logit:

```r
dat_lin <- data.frame(y = dat$y, d = dat$d, dat[, x_names])
fwl <- est_dml(dat_lin, y = "y", d = "d", x = x_names, cross_fit = FALSE,
               learner_l = lrn("regr.lm"), learner_m = lrn("regr.lm"))
c(dml_linear = fwl$estimate,
  ols        = coef(lm(y ~ ., data = dat_lin))[["d"]])
#> dml_linear        ols
#>  0.6199947  0.6199947
```

With cross-fitted forests it is the estimator of Section 1's fourth row:

```r
set.seed(1)
fit_plm <- est_dml(
  dat, y = "y", d = "d", x = x_names,
  learner_l = lrn("regr.ranger", num.trees = 100),
  learner_m = lrn("classif.ranger", predict_type = "prob", num.trees = 100),
  folds = 5, seed = 1
)
fit_plm
#> Double ML estimate (partially linear regression, theta)
#>   Score: partialling_out; solution: pooled across 5 fold(s); 1 repetition(s)
#>   Estimate:   0.9426
#>   Std. Error: 0.1003
#>   95% CI:     [0.7460, 1.1391]
#>   N:          500 (treated: 245, control: 255)
#>   Nuisance RMSE: l_hat =  1.2, m_hat = 0.491
```

What θ means when the model is wrong. Suppose the effect varies, $Y = g(0, X) + D\,\tau(X) + \varepsilon$, so the PLM is misspecified. The estimator still converges to the best linear predictor of $\tilde{Y}$ on $\tilde{D}$. For binary D that limit is

$$\theta^* = \frac{\mathbb{E}\big[m(X)\,(1-m(X))\,\tau(X)\big]}{\mathbb{E}\big[m(X)\,(1-m(X))\big]},$$

an average of $\tau(X)$ with **overlap weights** $m(X)(1-m(X))$. Units whose treatment is nearly determined by X get almost no weight. This is neither the ATE nor the ATT. With the true nuisances supplied, the partially linear estimator lands on the overlap-weighted average and the interactive estimator of the next subsection lands on the ATE:

```r
hte <- simulate_hte_dgp(n = 5000, seed = 2)
w_overlap <- with(hte, m_true * (1 - m_true))
c(
  ATE               = mean(hte$tau_true),
  overlap_weighted  = with(hte, sum(w_overlap * tau_true) / sum(w_overlap)),
  plm_estimate      = est_dml(hte, "y", "d", l_hat = "l_true", m_hat = "m_true")$estimate,
  irm_estimate      = est_dml(hte, "y", "d", model = "irm",
                          p_hat = "m_true", mu0_hat = "g0_true", mu1_hat = "g1_true")$estimate
)
#>              ATE overlap_weighted     plm_estimate     irm_estimate
#>         1.998015         1.551531         1.594433         2.001470
```

When it breaks.

- Effects are heterogeneous and the ATE or ATT is the question. Use the interactive model.
- $\mathbb{E}[\tilde{D}^2]$ is small: the covariates almost determine the treatment. The denominator is close to zero and the estimate is noisy. `est_dml()` reports this quantity under `diagnostics$identification`.
- $\hat{m}$ is poor. The bias is $-\theta_0\,\mathbb{E}[(\hat{m} - m_0)^2]/\mathbb{E}[\tilde{D}^2]$ at second order (Section 2.4); nothing protects against it.

3.2 Interactive model: the doubly robust score

Model.

$$Y = g(D, X) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D, X] = 0; \qquad D \in \{0, 1\}, \quad m(X) = \mathbb{P}(D = 1 \mid X).$$

No restriction on how D and X interact. Targets: the ATE $\theta = \mathbb{E}[g(1, X) - g(0, X)]$ and the ATT $\mathbb{E}[g(1, X) - g(0, X) \mid D = 1]$.

Three representations, one orthogonal. Section 03 wrote the ATE three ways. Only one of them can carry a machine learner.

Representation	$\theta =$	Nuisance	Derivative at the truth	Orthogonal?
Regression adjustment	$\mathbb{E}[g(1,X) - g(0,X)]$	g	$\mathbb{E}[\Delta(1,X) - \Delta(0,X)]$	no
Inverse propensity weighting	$\mathbb{E}[Y\,H(D,X)]$	m	$-\mathbb{E}[Y\,\Delta(X)\,(\frac{D}{m^2} + \frac{1-D}{(1-m)^2})]$	no
AIPW	$\mathbb{E}[g(1,X) - g(0,X) + H(D,X)(Y - g(D,X))]$	(g,m)	0 in both directions	yes

The AIPW representation is the other two glued together so that their first-order errors cancel (Section 2.2 and Appendix 9.1). Its remainder is the product of the two nuisance errors (Section 2.4): double robustness.

Scores. All three targets have scores linear in θ, $\psi = \psi^b - \psi^a\theta$, with $\bar{p} = \mathbb{P}(D=1)$ and $p_G = \mathbb{P}(G=1)$ for a group indicator G:

Target	$\psi^b(W;\eta)$	ψ^a
ATE	$g(1,X) - g(0,X) + H(D,X)\,(Y - g(D,X))$	1
ATT	$(D - (1-D)\,\frac{m(X)}{1-m(X)})\,(Y - g(0,X))/\bar{p}$	$D/\bar{p}$
GATE	$\frac{G}{p_G}\,\psi^b_{\text{ATE}}$	$\frac{G}{p_G}$

The ATT score needs only m and $g(0,\cdot)$: the treated arm is never imputed. The group score reweights the ATE score by group membership.

> **Algorithm (DML for the ATE in the interactive model).** 1. Split into K folds. 2. For each k, fit $\hat{g}_{[k]}(0,\cdot)$ on untreated units outside I_k, $\hat{g}_{[k]}(1,\cdot)$ on treated units outside I_k, and $\hat{m}_{[k]}$ on all units outside I_k; clip $\hat{m}$ to $[\epsilon, 1-\epsilon]$. 3. For $i \in I_k$ compute $\hat{\varphi}_i = \hat{g}(1,X_i) - \hat{g}(0,X_i) + \hat{H}_i\,(Y_i - \hat{g}(D_i,X_i))$. 4. $\hat{\theta} = \mathbb{E}_n[\hat{\varphi}]$ and $\widehat{\text{SE}} = \text{sd}(\hat{\varphi})/\sqrt{n}$.

In R. `est_dml(model = "irm")` runs the algorithm; `est_aipw()` is the same estimator with the Section 03 interface, and the two agree to the last digit on the same folds. On the Section 1 sample with forests:

```
learners_irm <- list(
  learner_p   = lrn("classif.ranger", predict_type = "prob", num.trees = 100),
  learner_mu0 = lrn("regr.ranger", num.trees = 100),
  learner_mu1 = lrn("regr.ranger", num.trees = 100)
)
set.seed(1)
fit_ate <- do.call(est_dml, c(list(dat, y = "y", d = "d", x = x_names,
                                   model = "irm", estimand = "ATE", seed = 1), learners_irm))
set.seed(1)
fit_att <- do.call(est_dml, c(list(dat, y = "y", d = "d", x = x_names,
                                   model = "irm", estimand = "ATT", seed = 1), learners_irm))
set.seed(1)
fit_aipw <- est_aipw(dat, y = "y", d = "d", x = x_names, seed = 1,
                     learner_p = learners_irm$learner_p,
                     learner_mu0 = learners_irm$learner_mu0,
                     learner_mu1 = learners_irm$learner_mu1)
rbind(
  ATE      = c(estimate = fit_ate$estimate,  std.error = fit_ate$std.error),
  ATT      = c(estimate = fit_att$estimate,  std.error = fit_att$std.error),
  est_aipw = c(estimate = fit_aipw$estimate, std.error = fit_aipw$std.error)
)
#>           estimate std.error
#> ATE      0.9794514 0.1171775
#> ATT      1.1395642 0.1374168
#> est_aipw 0.9794514 0.1171775
```

When it breaks.

- Weak overlap. H divides by $\hat{m}$ and $1-\hat{m}$. Clipping at ϵ caps the weights at $1/\epsilon$; trimming units outside $[\epsilon, 1-\epsilon]$ changes the target population. Report both (Section 03).

- Both nuisances poor. The product $\|\widehat{m} - m_0\| \, \|\widehat{g} - g_0\|$ must be $o(n^{-1/2})$; one accurate nuisance buys slack for the other, but not for both.
- Non-binary treatment. Use the partially linear model, or the generalizations in Section 8.

3.3 High-dimensional linear model: double lasso

Model.

$$Y = \alpha D + \beta' W + \varepsilon, \qquad D = \gamma' W + v,$$

with p controls, p comparable to or larger than n, and **approximate sparsity**: the sorted coefficients decay fast enough that a few hundred of them carry the signal. This is the partially linear model with linear nuisances and the lasso as the learner. It is where the DML story started (Belloni, Chernozhukov, and Hansen 2014).

The naive procedure and its failure. *Single selection*: run a lasso of Y on (D, W), keep the selected controls, refit by least squares, report the usual standard error. The lasso keeps controls that predict Y. It drops controls that predict D strongly but Y weakly. Those are exactly the confounders that omitted-variable bias is made of. In moment language the procedure solves $\mathbb{E}[(Y - \alpha D - \beta' W)\, D] = 0$ with an estimated β, and the derivative in the direction Δ is $-\mathbb{E}[WD]\,\Delta \neq 0$. Not orthogonal.

Double lasso. Partial out with the lasso on both sides.

1. Lasso Y on W; residuals $\check{Y}$.
2. Lasso D on W; residuals $\check{D}$.
3. Least squares of $\check{Y}$ on $\check{D}$; robust standard error.

Controls that matter for *either* equation survive in at least one residual. That is the orthogonality of Section 3.1 again. *Double selection* (least squares on D and the union of the two selected sets) is an equivalent variant.

Cross-fitting or not. With the theoretical plug-in penalty, sparsity theory guarantees that the lasso does not overfit, and no sample splitting is needed (Theorem 4.2.1 in the book). With a cross-validated penalty that guarantee is gone, so cross-fit. There is a second subtlety. The lasso *shrinks* the coefficients it keeps. Shrinkage is nuisance error like any other, and Section 2.4 says it is amplified by $1/\mathbb{E}[\tilde{D}^2]$. Post-selection refitting removes it; using the shrunken lasso predictions as $\widehat{\ell}$ and $\widehat{m}$ does not. The simulation shows both effects.

Simulation. Example 4.3.1 of the book: $p = n = 100$, $\beta_j = \gamma_j = 1/j^2$, $W \sim N(0, I_p)$, $v \sim N(0, 0.25^2)$, $\alpha = 1$. Three estimators.

Estimator	Steps
Single selection	lasso Y on (D, W); least squares on D and the selected W
Post-double-selection	lasso Y on W and D on W; least squares on D and the union of the selections
CV lasso, cross-fitted	`est_dml()` with cross-validated lasso *predictions* as $\widehat{\ell}$ and $\widehat{m}$

```
lasso_results <- run_double_lasso_experiment(
  n_sims = n_sims_lasso, n = 100, p = 100, theta = 1, sd_v = 0.25, folds = 5, seed = 1
)
lasso_tab <- summarise_dml_experiment(lasso_results)
pick_lasso <- function(estimator, column) lasso_tab[[column]][lasso_tab$estimator == estimator]
msq_dtilde <- mean(lasso_results$msq_treatment_residual, na.rm = TRUE)
msq_m_err  <- mean(lasso_results$msq_m_error, na.rm = TRUE)
```

```
plot_dml_experiment(lasso_results, colors = lasso_colors, ncol = 3)
```

Table 2: Lasso-based estimators, true effect 1, p = n = 100, residual treatment standard deviation 0.25.

Estimator	Bias	SD	RMSE	Mean reported SE	95% coverage
Single selection	0.931	0.099	0.937	0.091	0.00
Post-double-selection	-0.021	0.516	0.512	0.514	0.95
CV lasso, cross-fitted	0.744	0.361	0.826	0.356	0.45

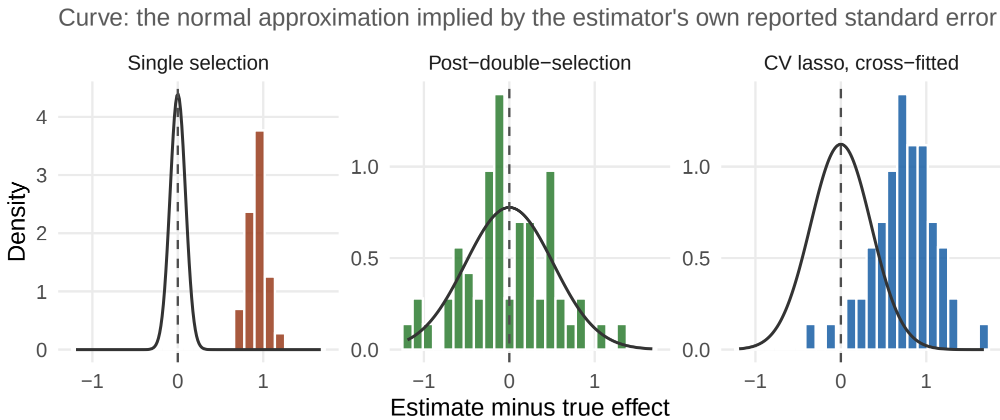

Figure 2: Single selection loses the confounders that predict the treatment. Post-double-selection keeps them and refits without shrinkage. Shrunken cross-validated lasso predictions fail here because the residual treatment variance is tiny.

Reading the results.

- **Single selection** has a bias of 0.93. The treatment is unpenalized and almost collinear with the leading controls, so the outcome lasso keeps D and drops most of those controls. Their effects, $\sum_j \beta_j \gamma_j$ over the dropped set, land on the coefficient of D (Appendix 9.1).
- **Post-double-selection** is centred (bias -0.02) with coverage 0.95. The lasso for D picks up the confounders the lasso for Y misses, and the least-squares refit carries no shrinkage.
- **Cross-validated lasso predictions**, cross-fitted, are biased (0.74) despite the orthogonal score and cross-fitting. The residual treatment variance in this design is $\mathbb{E}[\tilde{D}^2] \approx 0.10$, and the lasso's squared error in $\hat{m}$ is about 0.035. Both nuisances are shrunk toward zero in the same direction, so the remainder of Section 2.4, $(\mathbb{E}[(\hat{\ell} - \ell)(\hat{m} - m)] - \theta\, \mathbb{E}[(\hat{m} - m)^2])/\mathbb{E}[\tilde{D}^2]$, is positive and multiplied by roughly 10. This is the weak-residual-variation failure of Section 3.1 combined with shrinkage.

Three lessons. Refit after selecting when the lasso partials out; that is what post-lasso and the plug-in `rlasso` do. Read $\mathbb{E}[\tilde{D}^2]$ before trusting any partialling-out estimate; `est_dml()` reports it. And treat shrinkage as nuisance error: it counts against the rate condition like any other.

When it breaks.

- Sparsity fails: many small coefficients that together matter. The lasso cannot approximate either nuisance, and no penalty choice helps. Move to nonlinear learners or richer dictionaries.
- Cross-validated penalties without cross-fitting: the overfitting bias of Section 1 returns.
- Perfect or near-perfect prediction of D by W: the residual $\tilde{D}$ has little variation, and every nuisance error is amplified.

4 Cross-fitting

4.1 The problem it solves

Section 2.3 left one term open, the empirical process term

$$(b) = \sqrt{n}\,(\mathbb{E}_n - \mathbb{E})\big[\psi(\theta_0, \hat{\eta}) - \psi(\theta_0, \eta_0)\big].$$

It is a centred average, but of a function that was chosen by looking at the same data. There are two ways to make it vanish.

- **Restrict the learner.** If every function the learner can produce lies in a "simple" class (a Donsker class), a uniform central limit theorem applies and $(b) \to 0$. Parametric models and low-dimensional smooth nonparametrics qualify. Forests, boosting, neural networks, and the lasso with data-driven tuning do not.
- **Break the dependence.** Fit $\hat{\eta}$ on observations that are not used to evaluate the score. Then, conditional on the fitting data, $\hat{\eta}$ is a fixed function and (b) is an ordinary centred average.

Cross-fitting takes the second route. It places no restriction on the learner. Section 1's row "Orthogonal score, full-sample forests" shows what happens when neither route is taken: the score is right, the learner is adaptive, and (b) does not vanish.

4.2 The construction

> **Algorithm (K-fold cross-fitting).** 1. Split the indices $\{1, \dots, n\}$ into K folds $I_1, \dots, I_K$ of similar size, stratified on D when D is binary. 2. For each k, fit the nuisances $\hat{\eta}_{[k]}$ on the complement I_k^c. 3. For each $i \in I_k$, evaluate the score with $\hat{\eta}_{[k(i)]}$, the fit that did not use observation i. 4. Solve $\mathbb{E}_n[\psi(W_i; \theta, \hat{\eta}_{[k(i)]})] = 0$ on the pooled scores (or fold by fold; Section 5).

Every observation contributes a score computed from a model that never saw it. Every nuisance fit uses a fraction $(K-1)/K$ of the data. With $K = 2$ this is the original "swap" scheme; $K = 5$ is the usual default; larger K helps in small samples (Remark 9.4.5 in the book).

4.3 Why it works: the conditional argument

Fix a fold k and condition on its training data $T_k = \{W_i : i \notin I_k\}$. The fitted nuisance $\hat{\eta}_{[k]}$ is then a fixed function. For $i \in I_k$ define

$$Z_i = \psi(W_i; \theta_0, \hat{\eta}_{[k]}) - \psi(W_i; \theta_0, \eta_0).$$

Conditional on T_k the Z_i are i.i.d., with mean $\mu_k = \mathbb{E}[Z_i \mid T_k]$ and variance $\sigma_k^2 = \mathbb{E}[Z_i^2 \mid T_k] - \mu_k^2$. The fold's contribution to (b) is $\sqrt{n_k}\,(\bar{Z}_k - \mu_k)$, and by Chebyshev's inequality

$$\mathbb{P}\big(|\sqrt{n_k}\,(\bar{Z}_k - \mu_k)| > t \,\big|\, T_k\big) \;\le\; \frac{\sigma_k^2}{t^2}.$$

The variance σ_k^2 is bounded by $\mathbb{E}[(\psi(\hat{\eta}_{[k]}) - \psi(\eta_0))^2 \mid T_k]$, which goes to zero whenever $\hat{\eta}_{[k]}$ is consistent and the score is smooth in η. So each fold's centred average is $o_p(1)$ *after* the $\sqrt{n}$ scaling, and summing over K folds gives $(b) = o_p(1)$. The means μ_k are exactly the bias term (c) of Section 2.3, which orthogonality handles. Appendix 9.2 writes this out.

Two features of the argument matter. Nothing in it depends on what the learner is. And the only requirement on $\hat{\eta}$ is consistency, which is weaker than the rate condition of Step 2; the rate is needed for (c), not for (b).

4.4 What it costs, and what it does not fix

Costs.

- K nuisance fits instead of one, on smaller samples. With adaptive learners this is the price of valid inference, not a nuisance.

- The estimate depends on the random partition. Section 5 removes that dependence with repeated partitions.

What it does not fix. Cross-fitting cures the own-observation bias. It does nothing about the choice of moment. Table 1 has the proof: the cross-fitted plug-in forest keeps essentially the full bias of the in-sample plug-in. Sample splitting with a non-orthogonal score is the "natural algorithm, draft 3" of the Stanford lectures, and it still fails. Nor does cross-fitting create identification, repair overlap, or shrink the rate condition.

In R. The residual diagnostics of `est_dml()` make the own-observation bias visible on a single sample. In-sample forests produce residuals that are too small, because the fitted values have absorbed part of the noise.

```
set.seed(1)
fit_insample <- est_dml(
  dat, y = "y", d = "d", x = x_names,
  learner_l = lrn("regr.ranger", num.trees = 100),
  learner_m = lrn("classif.ranger", predict_type = "prob", num.trees = 100),
  cross_fit = FALSE
)
compare <- function(fit) {
  q <- fit$diagnostics$nuisance
  c(rmse_l = q$rmse[q$nuisance == "l_hat"],
    rmse_m = q$rmse[q$nuisance == "m_hat"],
    mean_sq_D_resid = fit$diagnostics$identification$mean_sq_treatment_residual,
    estimate = fit$estimate)
}
round(rbind(full_sample = compare(fit_insample), cross_fitted = compare(fit_plm)), 3)
#>              rmse_l rmse_m mean_sq_D_resid estimate
#> full_sample   0.523  0.264           0.069    0.740
#> cross_fitted  1.204  0.491           0.241    0.943
```

The in-sample fit looks better on every nuisance metric. That is the warning sign, not a success: residuals that small are impossible for a treatment with this much unexplained variation. Cross-fitted residual metrics are honest estimates of the out-of-sample nuisance error, and the rate condition of Section 2.4 is about that error.

5 The recipe in practice

5.1 The generic algorithm

Inputs. Data $\{W_i\}_{i=1}^n$; an orthogonal score $\psi(W;\theta,\eta)$ that identifies θ_0; a learner for each component of η; the number of folds K; the number of repeated partitions S.

For each partition $s = 1, \dots, S$**:** 1. Draw K folds. 2. Fit the nuisances on each complement; predict on the held-out fold. 3. Form the cross-fitted scores $\psi(W_i;\theta,\hat\eta_{[k(i)]})$. 4. Solve $\mathbb{E}_n[\psi] = 0$ for $\hat\theta_s$; compute $\hat\sigma_s^2$.

Report $\hat\theta_1, \hat\sigma_1$ if $S = 1$, otherwise the median aggregates of Section 5.4.

`est_dml()` is this algorithm. The rest of the section is about the choices inside it.

5.2 Linear scores and the Jacobian

Every score in this lecture is linear in θ:

$$\psi(W;\theta,\eta) = \psi^b(W;\eta) - \psi^a(W;\eta)\,\theta.$$

Then step 4 has a closed form and so does the variance:

$$\hat\theta = \frac{\sum_i \psi_i^b}{\sum_i \psi_i^a}, \qquad \hat J = \frac{1}{n}\sum_i \psi_i^a, \qquad \hat\sigma^2 = \frac{1}{\hat J^2}\cdot\frac{1}{n-1}\sum_i\left(\psi_i^b - \psi_i^a\,\hat\theta\right)^2, \qquad \widehat{\mathrm{SE}} = \hat\sigma/\sqrt{n}.$$

Score	ψ^a	ψ^b	$\hat{J}$ estimates
Partialling out	$\tilde{D}^2$	$\tilde{Y}\tilde{D}$	$\mathbb{E}[\tilde{D}^2]$, the residual treatment variance
ATE (AIPW)	1	$g(1,X) - g(0,X) + H(D,X)(Y - g(D,X))$	1
ATT	$D/\bar{p}$	$(D - (1-D)\frac{m}{1-m})(Y - g(0,X))/\bar{p}$	1

The Jacobian is where identification strength lives. For the ATE and ATT it is one by construction. For the partially linear model it is the residual treatment variance, and a small value inflates both the estimate and every remainder term (Sections 3.1 and 3.3).

5.3 Pooled or fold-averaged solutions

Step 4 can be run in two ways.

- **Pooled** (`solve = "pooled"`, DML2 in the paper): stack the cross-fitted scores of all folds and solve once.
- **Fold average** (`solve = "fold_average"`, DML1): solve within each fold, then average the K solutions.

Both use the variance formula above with the pooled scores. They agree asymptotically. In finite samples the pooled solution is more stable, because a fold with a small $\hat{J}_k$ does not get a full $1/K$ share of the answer. Pooled is the default. The fold-level solutions are still worth reading: their spread is a cheap diagnostic of how much the answer depends on the partition.

```
set.seed(1)
fit_fold_avg <- est_dml(
  dat, y = "y", d = "d", x = x_names,
  learner_l = lrn("regr.ranger", num.trees = 100),
  learner_m = lrn("classif.ranger", predict_type = "prob", num.trees = 100),
  folds = 5, seed = 1, solve = "fold_average"
)
rbind(pooled = c(estimate = fit_plm$estimate, std.error = fit_plm$std.error),
      fold_average = c(estimate = fit_fold_avg$estimate, std.error = fit_fold_avg$std.error))
#>               estimate std.error
#> pooled       0.9425729 0.1002728
#> fold_average 0.9417870 0.1002750
fit_plm$fold_estimates
#>   rep fold  estimate   n
#> 1   1    1 1.2772629 100
#> 2   1    2 0.8150658 100
#> 3   1    3 0.7147882 100
#> 4   1    4 0.9720484 100
#> 5   1    5 0.9297697 100
```

5.4 Repeated cross-fitting

One partition is one random draw. Its estimate carries partition noise on top of sampling noise. Repeating the whole procedure on S independent partitions and aggregating removes the dependence on any one of them. Chernozhukov et al. (2018, Section 3.4) recommend the median:

$$\tilde{\theta} = \text{median}_s\ \hat{\theta}_s, \qquad \tilde{\sigma}^2 = \text{median}_s\Big(\hat{\sigma}_s^2 + (\hat{\theta}_s - \tilde{\theta})^2\Big).$$

The second term inside the median charges the variance for the spread across partitions, so the interval does not pretend the partition was fixed. The mean works too, with the same correction; the median is less sensitive to one unlucky split.

```
set.seed(1)
fit_rep <- est_dml(
  dat, y = "y", d = "d", x = x_names,
  learner_l = lrn("regr.ranger", num.trees = 100),
  learner_m = lrn("classif.ranger", predict_type = "prob", num.trees = 100),
  folds = 5, n_rep = 5, seed = 1
```

```
)
fit_rep$repetitions
#>   rep  estimate std.error   n
#> 1   1 0.9425729 0.1002728 500
#> 2   2 0.9710615 0.1005475 500
#> 3   3 0.9829228 0.1014475 500
#> 4   4 0.9233411 0.1010419 500
#> 5   5 0.9615799 0.1019931 500
c(estimate = fit_rep$estimate, std.error = fit_rep$std.error)
#>  estimate std.error
#> 0.9615799 0.1020583
```

Use $S \geq 5$ whenever the learners are flexible and n is moderate. With $S = 1$, two analysts with different seeds report different numbers from the same data.

5.5 Choosing and tuning the learners

The rate condition is about the *error* of each nuisance estimate. The cross-fitted residuals measure that error out of sample, so they also rank learners.

> **Selecting learners to minimise the DML bias.** For each candidate learner j and each nuisance, compute the cross-fitted mean squared prediction error, for instance $\mathbb{E}_n[\check{Y}_j^2]$ and $\mathbb{E}_n[\check{D}_j^2]$. Pick the learner with the smallest error for each nuisance separately. Or form a linear ensemble of the candidates with the same criterion.

Three rules go with the box.

- **Few candidates.** Choosing among a handful of learners does not disturb the theory (Corollary 9.2.3 in the book). Searching over hundreds does; the validation data are then being used to fit.
- **Tune inside the training folds.** Cross-validation for hyperparameters must happen on the $K-1$ training folds, never on the held-out fold. Data-driven tuning plus cross-fitting is fine; data-driven tuning without cross-fitting is Section 1's overfitting bias.
- **Different learners for different nuisances.** Nothing requires the same method for ℓ and m. A binary treatment often wants a calibrated classifier for m and a flexible regression for ℓ.

On the Section 1 sample, three candidates for the partially linear nuisances: linear models, forests, and a lasso on the covariates and their squares.

```
dat_sq <- dat
sq_names <- paste0(x_names, "_sq")
dat_sq[sq_names] <- dat[x_names]^2

candidates <- list(
  linear = list(x = x_names,
                l = lrn("regr.lm"),
                m = lrn("classif.log_reg", predict_type = "prob")),
  forest = list(x = x_names,
                l = lrn("regr.ranger", num.trees = 100),
                m = lrn("classif.ranger", predict_type = "prob", num.trees = 100)),
  lasso_squares = list(x = c(x_names, sq_names),
                       l = lrn("regr.cv_glmnet", nfolds = 5),
                       m = lrn("classif.cv_glmnet", predict_type = "prob", nfolds = 5))
)
compare_learners <- t(sapply(candidates, function(cand) {
  set.seed(1)
  fit <- est_dml(dat_sq, y = "y", d = "d", x = cand$x,
                 learner_l = cand$l, learner_m = cand$m, folds = 5, seed = 1)
  q <- fit$diagnostics$nuisance
  c(rmse_l = q$rmse[q$nuisance == "l_hat"], rmse_m = q$rmse[q$nuisance == "m_hat"],
    estimate = fit$estimate, std.error = fit$std.error)
}))
round(compare_learners, 3)
#>               rmse_l rmse_m estimate std.error
#> linear         1.341  0.487    0.606     0.125
```

```
#> forest        1.204  0.491    0.943     0.100
#> lasso_squares  1.182  0.474    0.972     0.102
```

Read the table by column, not by row. The learner with the smallest `rmse_l` is the right choice for ℓ, the one with the smallest `rmse_m` for m, and they need not be the same. Here the squares give the lasso the curvature it needs, and it edges out the forest on both nuisances. The two flexible candidates agree on the effect. The linear candidate, with the largest errors, is the outlier.

5.6 Folds, clipping, and trimming

- **Folds.** $K = 5$ is a sensible default. Use $K \geq 10$ when n is a few hundred, so that each nuisance fit sees enough data. Stratify the folds on D when D is binary; `est_dml()` does this automatically.
- **Clipping.** In the interactive model, cap $\hat{m}$ to $[\epsilon, 1-\epsilon]$ before forming H. The default $\epsilon = 0.01$ bounds the weights at 100. Clipping is a variance device and leaves the estimand alone.
- **Trimming.** Dropping units with $\hat{m}$ outside $[\epsilon, 1-\epsilon]$ changes the population the estimate refers to, as in Section 03. Report the trimmed share and, when possible, the estimate with and without trimming.
- **Binary treatment in the partially linear model.** Predict D with a classifier and use its probability as $\hat{m}$; a regression forest on a 0/1 outcome works but is less well calibrated in the tails.

6 Diagnostics and failure modes

6.1 What to report

A DML estimate is only as credible as the nuisances behind it. Report the quantities below with every estimate; `est_dml()` returns all of them.

Diagnostic	What it checks	Where
Cross-fitted RMSE and R^2 of each nuisance	rate condition; learner choice	`diagnostics$nuisance`
Residual treatment variance $\mathbb{E}_n[\tilde{D}^2]$	identification strength (partially linear)	`diagnostics$identification`
Propensity range, clipped share, effective sample sizes	overlap (interactive)	`diagnostics$propensity`, `diagnostics$weights`
Fold-level estimates	dependence on one partition	`fold_estimates`
Repetition estimates	dependence across partitions	`repetitions`
Estimates across two or three learners refsec-learners	model dependence	rerun; Section
In-sample versus cross-fitted residuals refsec-crossfit	overfitting of external predictions	`cross_fit = FALSE`; Section
Placebo outcomes, sensitivity analysis	identification	outside the estimator; Section 03

```
fit_plm$diagnostics$nuisance
#>   nuisance   target      rmse         r2
#> 1    l_hat E[Y | X] 1.2036677 0.36908202
#> 2    m_hat E[D | X] 0.4910754 0.03692374
fit_plm$diagnostics$identification$mean_sq_treatment_residual
#> [1] 0.2411551
fit_ate$diagnostics$propensity$clipped_summary
#>       min       q01       q05    median      mean       q95       q99       max
#> 0.1485794 0.1776209 0.2565770 0.4868929 0.4941691 0.7573365 0.8097275 0.9292500
c(ess_treated = fit_ate$diagnostics$weights$ess_treated,
  ess_control = fit_ate$diagnostics$weights$ess_control)
#> ess_treated ess_control
#>    211.8232    196.6716
```

6.2 Failure modes

Each entry: the symptom, then the remedy.

- **Weak residual variation** (partially linear). $\mathbb{E}_n[\tilde{D}^2]$ is a small fraction of $\mathrm{var}(D)$; the standard error is large; fold estimates scatter. Remedy: none inside the estimator. The covariates nearly determine the treatment. Accept the imprecision, or reconsider what variation identifies the effect.
- **Weak overlap** (interactive). $\hat{m}$ near 0 or 1 for many units; effective sample sizes far below n_1 and n_0. Remedy: clip, trim and report the trimmed share, or target the ATT if the untreated support is the problem.
- **Nuisances too poor.** Large cross-fitted RMSE; the residual bias of Section 2.4 is not negligible. Remedy: better learners chosen by cross-fitted error, more data, or a doubly robust score when one nuisance is much easier than the other.
- **Shrinkage with weak variation.** The lasso case of Section 3.3. Remedy: refit after selection.
- **Partition dependence.** Fold estimates or repetitions disagree by more than a standard error. Remedy: repeated cross-fitting with median aggregation; larger K in small samples.
- **In-sample nuisances from outside.** Residual metrics look too good; the estimate drifts toward the plug-in. Remedy: the contract for external predictions is out-of-fold, and the function cannot verify it. Check the fold structure of what you were handed.
- **Heterogeneous effects read as an ATE.** The partially linear coefficient is an overlap-weighted average (Section 3.1). Remedy: the interactive model for a binary treatment.
- **Bad controls and hidden confounding.** No DML diagnostic detects them. Remedy: the placebo and sensitivity tools of Section 03, and the adjustment-set discipline of Section 03.
- **Too few observations per fold.** Nuisance fits fail or degrade. Remedy: $K \geq 10$, simpler learners, repeated splits.

The common thread: DML moves the difficulty from specifying a functional form to learning nuisances well and saying how well they were learned. The diagnostics are the second half of that bargain.

7 Practice

7.1 The replications

Two papers anchor this section. Their replication notebooks are planned for `inst/replications/04_Doubly_Robust_and_Dou` and are not yet part of the package.

Chernozhukov et al. (2018), Tables 1 and 2.

- *Table 1, Pennsylvania reemployment bonus experiment.* The treatment is a randomized bonus offer; the outcome is unemployment duration. Because assignment is randomized, identification is not the question. The table teaches two things: DML with several learners agrees with the experimental benchmark, and the partially linear and interactive models give the same answer when the effect is close to constant.
- *Table 2, 401(k) eligibility and net financial assets.* Eligibility is observational; the design argues that income and a few demographics make it as good as random. The table compares lasso, trees, forests, boosting, and an ensemble in both models, with repeated splits. Watch the cross-fitted RMSE rows next to the estimates: the learners that predict better move together, and the estimate is far below the raw difference in means.

Ellickson, Kar, and Reeder (2023), targeted promotions. Emails are targeted by a rule that depends on customer history, so raw comparisons across emails are confounded. Stage one builds an orthogonal (AIPW) score for each customer and each email-versus-baseline contrast with forest nuisances; those are `est_dml(model = "irm")$score` plus the estimate, one number per unit. Stage two regresses the scores on the components that distinguish the emails. The replication teaches that DML scores are data: once orthogonalized, they can be projected, averaged within groups, or fed to a policy rule, and the projection inherits valid inference.

7.2 A reporting checklist

1. State the estimand and the model: a partially linear coefficient, an ATE, or an ATT.
2. Defend the adjustment set as pre-treatment and causally valid (Section 03).
3. Name the learner for each nuisance and how it was chosen; report the cross-fitted RMSE next to the estimate.
4. Report the residual treatment variance (partially linear) or the propensity range, clipped share, and effective sample sizes (interactive).
5. State K, the number of repeated partitions, the solution method, and any clipping or trimming.
6. Show the fold-level and repetition-level spread.
7. Compare two or three learners; a large gap is model dependence.
8. If external predictions were used, state that they are out-of-fold and how the folds were built.
9. Run a placebo outcome or a sensitivity analysis. DML does not deliver identification.
10. Interpret a partially linear coefficient as an overlap-weighted effect when heterogeneity is plausible.

8 Extensions

The recipe does not change across designs. Only the score does. Each item below is one orthogonal score and one pair of nuisances.

- **Instrumental variables** (Section 05). Partially linear IV residualizes Y, D, and the instrument Z on X, then runs two-stage least squares on the residuals. The interactive version ratios two AIPW-type scores, one for Y and one for D, with the instrument's propensity in the denominators, and targets the LATE.
- **Difference-in-differences** (Section 07). The doubly robust DiD score applies the AIPW construction to the outcome *change*, with a propensity for belonging to the treated group. Conditional parallel trends replaces conditional ignorability.
- **Heterogeneous effects** (Section 08). The residuals of the partially linear model satisfy $\tilde{Y} \approx \tau(X)\tilde{D}$; the R-learner fits $\tau(X)$ by minimizing $(\tilde{Y} - \tau(X)\tilde{D})^2$. The DR-learner regresses the AIPW score on X. Both reuse this lecture's nuisances.
- **Group and treated-population effects.** The ATT and group-ATE scores of Section 3.2; one estimator per group, same folds.
- **Automatic DML.** For any linear functional of the regression function there is a weight $\alpha(W)$, the Riesz representer, that plays the role of $H(D, X)$ in the AIPW score. Automatic DML learns α by machine learning instead of deriving it by hand, which extends the recipe to functionals for which no closed form exists.
- **Targeted maximum likelihood.** TMLE debiases the plug-in by refitting $\hat{g}$ along the direction $H(D, X)$ until the score has sample mean zero, then plugs in. It is first-order equivalent to DML with the same nuisances and also needs cross-fitting with adaptive learners.

9 Appendix

9.1 Orthogonality calculations

Notation. $W = (Y, D, X)$. The nuisance is η, its true value is η_0, and $\Delta = \eta - \eta_0$ is a fixed direction. The moment is $M(\theta, \eta) = \mathbb{E}[\psi(W; \theta, \eta)]$ and

$$\partial_\eta M(\theta_0, \eta_0)[\Delta] = \frac{d}{dt} M(\theta_0, \eta_0 + t\Delta)\Big|_{t=0}.$$

$\|f\| = \sqrt{\mathbb{E}[f(W)^2]}$ is the L^2 norm. Two facts do all the work. Under $\mathbb{E}[\varepsilon \mid D, X] = 0$, the outcome error times any function of (D, X) has mean zero. Under $\mathbb{E}[D \mid X] = m_0(X)$, the treatment residual $\tilde{D} = D - m_0(X)$ times any function of X has mean zero.

9.1.1 Plug-in score

Score $\psi = g(1,X) - g(0,X) - \theta$, nuisance g. With $g = g_0 + t\Delta$,

$$M(\theta_0, g_0 + t\Delta) = t\,\mathbb{E}[\Delta(1,X) - \Delta(0,X)], \qquad \partial_g M[\Delta] = \mathbb{E}[\Delta(1,X) - \Delta(0,X)].$$

The derivative is not zero. The moment is linear in Δ, so there is no second-order term to fall back on. The bias of the plug-in is exactly $\mathbb{E}[\hat{g}(1,X) - g_0(1,X)] - \mathbb{E}[\hat{g}(0,X) - g_0(0,X)]$. Whatever regularization does to the treatment contrast passes straight through.

9.1.2 Partialling-out score

Score $\psi = (Y - \ell(X) - \theta\,(D - m(X)))(D - m(X))$, nuisance $\eta = (\ell, m)$. Write $U = Y - \ell_0(X) - \theta_0 \tilde{D}$. In the partially linear model $U = \varepsilon$, so $\mathbb{E}[U \mid D, X] = 0$.

Direction in ℓ. With $\ell = \ell_0 + t\Delta_\ell$ and $m = m_0$,

$$\psi = (U - t\Delta_\ell(X))\tilde{D}, \qquad \partial_\ell M[\Delta_\ell] = -\mathbb{E}[\Delta_\ell(X)\,\tilde{D}] = -\mathbb{E}[\Delta_\ell(X)\,\mathbb{E}[\tilde{D} \mid X]] = 0.$$

Direction in m. With $m = m_0 + t\Delta_m$ and $\ell = \ell_0$, $D - m = \tilde{D} - t\Delta_m$ and

$$\psi = (U + t\theta_0\Delta_m)(\tilde{D} - t\Delta_m) = U\tilde{D} - t\,U\Delta_m + t\,\theta_0\Delta_m\tilde{D} - t^2\theta_0\Delta_m^2.$$

The derivative at $t = 0$ is $-\mathbb{E}[U\Delta_m(X)] + \theta_0\,\mathbb{E}[\Delta_m(X)\tilde{D}] = 0 + 0$.

Exact remainder. Put $\Delta_\ell = \hat{\ell} - \ell_0$ and $\Delta_m = \hat{m} - m_0$, both fixed functions. Then $Y - \hat{\ell} - \theta_0(D - \hat{m}) = U - \Delta_\ell + \theta_0\Delta_m$ and $D - \hat{m} = \tilde{D} - \Delta_m$, so

$$\psi(W; \theta_0, \hat{\eta}) = (U - \Delta_\ell + \theta_0\Delta_m)(\tilde{D} - \Delta_m).$$

Take expectations. The terms $U\tilde{D}$, $U\Delta_m$, $\Delta_\ell\tilde{D}$ and $\Delta_m\tilde{D}$ vanish by the two facts. What remains is

$$\mathbb{E}[\psi(W; \theta_0, \hat{\eta})] = \mathbb{E}[\Delta_\ell\,\Delta_m] - \theta_0\,\mathbb{E}[\Delta_m^2].$$

The expansion is exact: the score is bilinear in the nuisance errors, so there are no higher-order terms. Cauchy-Schwarz bounds it by $\|\Delta_\ell\|\,\|\Delta_m\| + |\theta_0|\,\|\Delta_m\|^2$. The Jacobian at $\hat{\eta}$ is $\mathbb{E}[(D - \hat{m})^2] = \mathbb{E}[\tilde{D}^2] + \|\Delta_m\|^2$, so with $\hat{\eta}$ held fixed the population solution is

$$\theta(\hat{\eta}) - \theta_0 = \frac{\mathbb{E}[\Delta_\ell\Delta_m] - \theta_0\,\mathbb{E}[\Delta_m^2]}{\mathbb{E}[\tilde{D}^2] + \mathbb{E}[\Delta_m^2]}.$$

Section 2.4 evaluates the second term of the numerator on the fitted propensity. Section 3.3 is this formula with shrinkage in both errors.

9.1.3 AIPW score for the ATE

Score $\psi = g(1,X) - g(0,X) + H(D,X)\,(Y - g(D,X)) - \theta$ with $H(D,X) = D/m(X) - (1-D)/(1-m(X))$, nuisance $\eta = (g, m)$. H_0 is H at m_0.

Inverse-propensity identity. For any function $f(D,X)$,

$$\mathbb{E}\Big[\frac{D}{m_0(X)}\,f(D,X)\Big] = \mathbb{E}\Big[\frac{\mathbb{E}[D \mid X]}{m_0(X)}\,f(1,X)\Big] = \mathbb{E}[f(1,X)], \qquad \mathbb{E}\Big[\frac{1-D}{1-m_0(X)}\,f(D,X)\Big] = \mathbb{E}[f(0,X)].$$

Direction in g. With $g = g_0 + t\Delta_g$ and $m = m_0$,

$$\partial_g M[\Delta_g] = \mathbb{E}[\Delta_g(1,X) - \Delta_g(0,X)] - \mathbb{E}[H_0(D,X)\,\Delta_g(D,X)] = 0$$

by the identity with $f = \Delta_g$.

Direction in m. Only H depends on m. With $m = m_0 + t\Delta_m$,

$$\dot H(D,X) := \frac{d}{dt} H\Big|_{t=0} = -\Delta_m(X)\Big(\frac{D}{m_0(X)^2} + \frac{1-D}{(1-m_0(X))^2}\Big), \qquad \partial_m M[\Delta_m] = \mathbb{E}[\dot H(D,X)\,(Y - g_0(D,X))] = 0,$$

because $\dot H$ is a function of (D,X) and $\mathbb{E}[Y - g_0(D,X) \mid D,X] = 0$.

Inverse propensity weighting alone. The score $YH(D,X) - \theta$ has the same derivative with Y in place of $Y - g_0(D,X)$: $\mathbb{E}[\dot H\, Y] = \mathbb{E}[\dot H\, g_0(D,X)] \neq 0$. Subtracting g_0 inside the weighted term is what makes the derivative vanish. That is the middle row of the table in Section 3.2.

Exact remainder. With $\Delta_g = \hat g - g_0$, $\Delta_m = \hat m - m_0$ and $\hat H$ equal to H at $\hat m$,

$$\mathbb{E}[\psi(W;\theta_0,\hat\eta)] = \mathbb{E}[\Delta_g(1,X) - \Delta_g(0,X)] + \underbrace{\mathbb{E}[\hat H\,(Y - g_0(D,X))]}_{=0} - \mathbb{E}[\hat H\,\Delta_g(D,X)].$$

Conditioning on X in the last term gives $\mathbb{E}[\frac{m_0}{\hat m}\Delta_g(1,X)] - \mathbb{E}[\frac{1-m_0}{1-\hat m}\Delta_g(0,X)]$. Collecting terms,

$$\mathbb{E}[\psi(W;\theta_0,\hat\eta)] = \mathbb{E}\Big[\frac{\hat m - m_0}{\hat m}\,\Delta_g(1,X)\Big] + \mathbb{E}\Big[\frac{\hat m - m_0}{1-\hat m}\,\Delta_g(0,X)\Big].$$

Again the expansion is exact. If $\hat m$ is clipped to $[\epsilon, 1-\epsilon]$, Cauchy-Schwarz gives

$$\big|\mathbb{E}[\psi(W;\theta_0,\hat\eta)]\big| \le \frac{1}{\epsilon}\,\|\hat m - m_0\|\,(\|\hat g(1,\cdot) - g_0(1,\cdot)\| + \|\hat g(0,\cdot) - g_0(0,\cdot)\|).$$

The product form is double robustness. The factor $1/\epsilon$ is the theoretical price of weak overlap, and the reason `est_dml()` clips.

9.1.4 ATT score

Score $\psi = \psi^b - \psi^a\theta$ with

$$\psi^b = \Big(D - (1-D)\frac{m(X)}{1-m(X)}\Big)\frac{Y - g(0,X)}{\bar p}, \qquad \psi^a = \frac{D}{\bar p},$$

nuisance $\eta = (g(0,\cdot), m, \bar p)$. At the truth, $\mathbb{E}[D\,(Y - g_0(0,X))] = \mathbb{E}[D\,(g_0(1,X) - g_0(0,X))] = \bar p\,\theta_0$ and $\mathbb{E}[(1-D)\frac{m_0}{1-m_0}(Y - g_0(0,X))] = 0$, so $M(\theta_0,\eta_0) = 0$.

Direction in $g(0,\cdot)$. With $g(0,\cdot) = g_0(0,\cdot) + t\Delta$,

$$\partial M[\Delta] = -\frac{1}{\bar p}\,\mathbb{E}\Big[\Big(D - (1-D)\frac{m_0}{1-m_0}\Big)\Delta(X)\Big] = -\frac{1}{\bar p}(\mathbb{E}[m_0\Delta] - \mathbb{E}[m_0\Delta]) = 0,$$

using $\mathbb{E}[D \mid X] = m_0$ and $\mathbb{E}[1 - D \mid X] = 1 - m_0$.

Direction in m. Only the odds $m/(1-m)$ depend on m. They multiply $(1-D)(Y - g_0(0,X))$, whose conditional mean given (D,X) is zero. The derivative is zero.

Direction in $\bar p$. The score is $\bar p^{-1}$ times a function that does not involve $\bar p$, so the derivative is $-M(\theta_0,\eta_0)/\bar p = 0$. In the estimator $\bar p$ cancels between numerator and denominator, so its estimation is immaterial.

Exact remainder. With $\Delta_g = \hat g(0,\cdot) - g_0(0,\cdot)$ and $\Delta_m = \hat m - m_0$,

$$\mathbb{E}[\psi(W;\theta_0,\hat\eta)] = \frac{1}{\bar p}\,\mathbb{E}\Big[\frac{\hat m - m_0}{1-\hat m}\,\Delta_g(X)\Big].$$

Only the untreated outcome model enters, and only in product with the propensity error. The ATT score is doubly robust in $(g(0,\cdot), m)$.

9.1.5 Single and double selection in the linear model

Model $Y = \alpha D + \beta' W + \varepsilon$ and $D = \gamma' W + v$ with $\mathbb{E}[\varepsilon \mid D, W] = 0$, $\mathbb{E}[v \mid W] = 0$, $\sigma_v^2 = \mathbb{E}[v^2]$, and $\Sigma_W = \mathbb{E}[WW']$. The coefficient of Y on W alone is $\pi = \alpha\gamma + \beta$.

Single selection is not orthogonal. The refit after selecting on the outcome equation solves $\mathbb{E}[(Y - \alpha D - \hat{\beta}' W)\, D] = 0$ in α, with β as the nuisance. In the direction Δ,

$$\partial_\beta M[\Delta] = -\mathbb{E}[D\, W']\,\Delta = -\gamma' \Sigma_W \Delta \neq 0$$

unless $\gamma = 0$, that is, unless there is no confounding.

One formula for both procedures. Let S be the set of controls kept by a selection and refit by least squares, with $\Sigma_W = I$. Then the partialling-out errors are $\Delta_{\ell,j} = -\pi_j$ and $\Delta_{m,j} = -\gamma_j$ for $j \notin S$, and zero otherwise. The exact formula of the partialling-out subsection gives

$$\theta(\hat{\eta}) - \alpha = \frac{\sum_{j\notin S} \pi_j \gamma_j - \alpha \sum_{j\notin S} \gamma_j^2}{\sigma_v^2 + \sum_{j\notin S} \gamma_j^2} = \frac{\sum_{j\notin S} \beta_j \gamma_j}{\sigma_v^2 + \sum_{j\notin S} \gamma_j^2},$$

which is also the limit of least squares of Y on D and W_S. It is omitted-variable bias, summed over the dropped controls.

- *Single selection* takes $S = S_\ell$, the controls the outcome lasso keeps. A control with small β_j and large γ_j is dropped and contributes its full $\beta_j \gamma_j$. When D is almost collinear with the leading controls, the lasso keeps D and drops those controls too; the numerator then contains their large products.
- *Post-double-selection* takes $S = S_\ell \cup S_m$. Every dropped control has small β_j and small γ_j, so each term of the numerator is a product of two small numbers. That is the remainder $\mathbb{E}[\Delta_\ell \Delta_m]$ in coordinates.
- *Shrunken predictions* keep the controls but shrink them: $\Delta_{m,j} = -s_j \gamma_j$ and $\Delta_{\ell,j} = -s_j' \pi_j$ with $s_j, s_j' \in (0,1]$ on the kept set. The same formula applies. The numerator no longer vanishes on the kept controls, and the denominator stays close to σ_v^2. When σ_v^2 is small, as in Table 2, the ratio is large.

9.2 Proof sketch of asymptotic normality

This appendix fills in Section 2.3 for a linear score with K-fold cross-fitting and the pooled solution. It follows Chernozhukov et al. (2018, Theorems 3.1 and 3.2), simplified to a scalar θ.

9.2.1 Setting

- $W_1, \dots, W_n$ are i.i.d. draws of W.
- The score is linear: $\psi(W; \theta, \eta) = \psi^b(W; \eta) - \psi^a(W; \eta)\,\theta$.
- The folds $I_1, \dots, I_K$ have $n_k = n/K$ observations each, and K is fixed.
- $\hat{\eta}_{[k]}$ is fitted on I_k^c. For $i \in I_k$ write $\psi_i^a = \psi^a(W_i; \hat{\eta}_{[k]})$ and $\psi_i^b = \psi^b(W_i; \hat{\eta}_{[k]})$.
- The estimator is $\hat{\theta} = \sum_i \psi_i^b / \sum_i \psi_i^a$ with $\hat{J} = \mathbb{E}_n[\psi_i^a]$.

For a fixed function η, $M(\theta, \eta) = \mathbb{E}[\psi(W; \theta, \eta)]$ is the expectation over a fresh draw of W. $\mathbb{E}_{n,k}$ is the average over $i \in I_k$. "Given I_k^c" means conditional on the observations outside fold k, which fixes $\hat{\eta}_{[k]}$.

9.2.2 Conditions

There are a set of functions $\mathcal{T}_n$ with $\mathbb{P}(\hat{\eta}_{[k]} \in \mathcal{T}_n) \to 1$ for every k, and sequences $r_n \to 0$ and $\lambda_n = o(n^{-1/2})$, such that for every $\eta \in \mathcal{T}_n$:

- **(C1) Identification.** $M(\theta_0, \eta_0) = 0$ and $J_0 = \mathbb{E}[\psi^a(W; \eta_0)] \neq 0$.
- **(C2) Orthogonality.** $\partial_\eta M(\theta_0, \eta_0)[\eta - \eta_0] = 0$.
- **(C3) Second-order remainder.** With $f(t) = M(\theta_0, \eta_0 + t(\eta - \eta_0))$, $\sup_{t \in (0,1)} |f''(t)| \leq \lambda_n$.
- **(C4) Consistency.** $\|\psi(\cdot; \theta_0, \eta) - \psi(\cdot; \theta_0, \eta_0)\| \leq r_n$ and $\|\psi^a(\cdot; \eta) - \psi^a(\cdot; \eta_0)\| \leq r_n$.
- **(C5) Moments.** $\sigma_0^2 = \mathbb{E}[\psi(W; \theta_0, \eta_0)^2]$ is finite and positive, and $\mathbb{E}[\psi^a(W; \eta_0)^2] < \infty$.

Appendix 9.1 gives (C2) and (C3) for the scores of this lecture: λ_n is of order $\|\hat{\ell}-\ell_0\|\,\|\hat{m}-m_0\|+\|\hat{m}-m_0\|^2$ for partialling out and $\epsilon^{-1}\|\hat{m}-m_0\|\,\|\hat{g}-g_0\|$ for AIPW. Learners whose L^2 error is $o(n^{-1/4})$ deliver $\lambda_n = o(n^{-1/2})$ and $r_n \to 0$.

Theorem. Under (C1) to (C5),

$$\sqrt{n}\,(\hat{\theta}-\theta_0) \to N(0,V), \qquad V = \sigma_0^2/J_0^2,$$

and $\hat{\sigma}^2 = \hat{J}^{-2}\,\mathbb{E}_n\big[(\psi_i^b - \psi_i^a\hat{\theta})^2\big] \to_p V$.

9.2.3 Proof

Step 0: representation. By construction $\sum_i(\psi_i^b - \psi_i^a\hat{\theta}) = 0$. Subtracting this from $\sum_i(\psi_i^b - \psi_i^a\theta_0)$ gives $\mathbb{E}_n[\psi(W_i;\theta_0,\hat{\eta}_{[k(i)]})] = (\hat{\theta}-\theta_0)\,\hat{J}$, so

$$\sqrt{n}\,(\hat{\theta}-\theta_0) = \frac{\sqrt{n}\,\mathbb{E}_n[\psi(W_i;\theta_0,\hat{\eta}_{[k(i)]})]}{\hat{J}}.$$

Step 1: a lemma for cross-fitted averages. Let $h(W;\eta)$ satisfy $\|h(\cdot;\eta)-h(\cdot;\eta_0)\| \le r_n$ for all $\eta \in \mathcal{T}_n$. Then

$$\mathbb{E}_{n,k}[h(W_i;\hat{\eta}_{[k]})] - \mathbb{E}_{n,k}[h(W_i;\eta_0)] = O_p(r_n).$$

Proof. Given I_k^c, the differences $Z_i = h(W_i;\hat{\eta}_{[k]}) - h(W_i;\eta_0)$, $i \in I_k$, are i.i.d. On the event $\hat{\eta}_{[k]} \in \mathcal{T}_n$, $\mathbb{E}[Z_i^2 \mid I_k^c] \le r_n^2$. By Jensen, $\bar{Z}_k^2 \le \mathbb{E}_{n,k}[Z_i^2]$, so $\mathbb{E}[\bar{Z}_k^2 \mid I_k^c] \le r_n^2$ and Markov's inequality gives $\mathbb{P}(|\bar{Z}_k| > t \mid I_k^c) \le r_n^2/t^2$. Take expectations and use $\mathbb{P}(\hat{\eta}_{[k]} \in \mathcal{T}_n) \to 1$. ☐

The conditioning is the point. Because $\hat{\eta}_{[k]}$ was fitted on I_k^c, it is a fixed function given I_k^c, and $\mathbb{E}[Z_i^2 \mid I_k^c]$ is an ordinary population moment at that function. Without cross-fitting $\hat{\eta}$ depends on W_i and no such bound is available.

Step 2: the Jacobian. Apply the lemma with $h = \psi^a$ and average over the K folds: $\hat{J} = \mathbb{E}_n[\psi^a(W_i;\eta_0)] + O_p(r_n) = J_0 + o_p(1)$ by the law of large numbers, (C4) and (C5).

Step 3: decomposition of the numerator. For each fold, with $Z_i = \psi(W_i;\theta_0,\hat{\eta}_{[k]}) - \psi(W_i;\theta_0,\eta_0)$,

$$\sqrt{n_k}\,\mathbb{E}_{n,k}[\psi(W_i;\theta_0,\hat{\eta}_{[k]})] = \underbrace{\sqrt{n_k}\,\mathbb{E}_{n,k}[\psi(W_i;\theta_0,\eta_0)]}_{(a_k)} + \underbrace{\sqrt{n_k}\,\big(\mathbb{E}_{n,k}[Z_i] - \mathbb{E}[Z_i \mid I_k^c]\big)}_{(b_k)} + \underbrace{\sqrt{n_k}\,\mathbb{E}[Z_i \mid I_k^c]}_{(c_k)}.$$

Since $n_k = n/K$,

$$\sqrt{n}\,\mathbb{E}_n[\psi(W_i;\theta_0,\hat{\eta}_{[k(i)]})] = \frac{1}{\sqrt{K}}\sum_{k=1}^{K}\big[(a_k)+(b_k)+(c_k)\big].$$

Step 4: the bias term (c_k). Let $f(t) = M(\theta_0, \eta_0 + t(\hat{\eta}_{[k]}-\eta_0))$. Then $\mathbb{E}[Z_i \mid I_k^c] = f(1) - f(0)$. By (C1), $f(0) = 0$. By (C2), $f'(0) = 0$. Taylor's theorem with the Lagrange remainder gives $f(1) = \frac{1}{2}f''(\bar{t})$ for some $\bar{t} \in (0,1)$, so by (C3)

$$|(c_k)| \le \tfrac{1}{2}\sqrt{n_k}\,\lambda_n = o(1).$$

Orthogonality removes the term of order $\sqrt{n}\,\|\hat{\eta}-\eta_0\|$. The rate condition removes the term of order $\sqrt{n}\,\|\hat{\eta}-\eta_0\|^2$.

Step 5: the empirical-process term (b_k). Given I_k^c, the Z_i are i.i.d. with variance at most $\mathbb{E}[Z_i^2 \mid I_k^c] \le r_n^2$ on the event $\hat{\eta}_{[k]} \in \mathcal{T}_n$. So (b_k) has conditional mean zero and conditional variance at most r_n^2, and Chebyshev's inequality gives

$$\mathbb{P}\big(|(b_k)| > t \,\big|\, I_k^c\big) \le r_n^2/t^2 \to 0.$$

Taking expectations, $(b_k) = o_p(1)$. This is the only step that uses cross-fitting, and it uses only consistency, not the rate.

Step 6: the oracle term and the limit. $K^{-1/2}\sum_k(a_k) = \sqrt{n}\,\mathbb{E}_n[\psi(W_i;\theta_0,\eta_0)]$ is $\sqrt{n}$ times an average of i.i.d. mean-zero variables with variance σ_0^2. By the central limit theorem it converges to $N(0,\sigma_0^2)$. Steps 0, 2, 4, 5 and Slutsky's lemma give

$$\sqrt{n}\,(\hat{\theta}-\theta_0) = \frac{N(0,\sigma_0^2)+o_p(1)}{J_0+o_p(1)} \;\to\; N(0,\sigma_0^2/J_0^2).$$

Step 7: the variance estimator. Write $\psi_i^b - \psi_i^a\hat{\theta} = \psi(W_i;\theta_0,\hat{\eta}_{[k(i)]}) - \psi_i^a(\hat{\theta}-\theta_0)$. The second piece has mean square $O_p(n^{-1})$ by Step 6 and (C5). For the first piece, the conditioning argument of Step 1 applied to Z_i^2 gives $\mathbb{E}_n[Z_i^2] = O_p(r_n^2)$, and the triangle inequality in the empirical L^2 norm gives

$$\left|\sqrt{\mathbb{E}_n[\psi(W_i;\theta_0,\hat{\eta}_{[k(i)]})^2]} - \sqrt{\mathbb{E}_n[\psi(W_i;\theta_0,\eta_0)^2]}\right| \le \sqrt{\mathbb{E}_n[Z_i^2]} = O_p(r_n).$$

So $\mathbb{E}_n[(\psi_i^b-\psi_i^a\hat{\theta})^2] \to_p \sigma_0^2$ by the law of large numbers, and $\hat{\sigma}^2 \to_p \sigma_0^2/J_0^2 = V$ by Step 2. □

9.2.4 Remarks

- **Where each ingredient enters.** Identification: Steps 2 and 4 ($f(0)=0$). Orthogonality: Step 4 ($f'(0)=0$). The rate: Step 4 ($\sqrt{n}\,\lambda_n \to 0$). Cross-fitting: Steps 1, 5 and 7, through the conditioning on I_k^c. Nuisance consistency: Steps 2, 5 and 7. Without cross-fitting, Step 5 needs a Donsker condition on the learner. Without orthogonality, (c_k) is of order $\sqrt{n}\,\|\hat{\eta}-\eta_0\|$, which diverges for any learner slower than $n^{-1/2}$.
- **Fold average.** The fold-average solution replaces Step 0 by $\hat{\theta}_k = \sum_{I_k}\psi_i^b/\sum_{I_k}\psi_i^a$ and averages the K solutions. The same steps apply fold by fold, with $\hat{J}_k \to_p J_0$, and the limit is unchanged.
- **Repeated partitions.** Steps 4 and 5 show that $\sqrt{n}\,(\hat{\theta}_s-\theta_0) = J_0^{-1}\sqrt{n}\,\mathbb{E}_n[\psi(W_i;\theta_0,\eta_0)] + o_p(1)$ for every partition s, and the leading term does not depend on the partition. So $\sqrt{n}\,(\hat{\theta}_s-\hat{\theta}_{s'}) = o_p(1)$, and the mean or median over a fixed number S of partitions has the same limit. The correction $(\hat{\theta}_s-\tilde{\theta})^2$ in the median variance is $o_p(n^{-1})$ in theory; it matters in finite samples, where the $o_p(1)$ terms are not yet small.

9.3 References

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